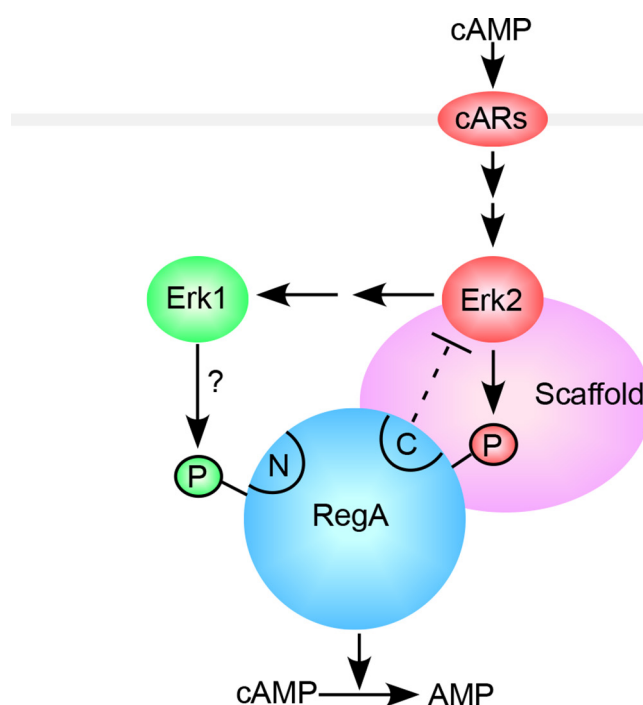


Mitogen-activated protein kinase regulation of the phosphodiesterase RegA in early *Dictyostelium* development

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Graphical Abstract

Model of MAPK regulation of phosphodiesterase RegA in *Dictyostelium discoideum*. Erk2 associates and downregulates RegA independently of the putative docking site. Erk1 possibly upregulates RegA through another phosphorylation site.

Abstract

Mitogen-activated protein kinase (MAPK) regulation of cAMP-specific phosphodiesterase function has been demonstrated in mammalian cells and suspected to occur in other eukaryotes. Epistasis analysis in the soil amoeba *Dictyostelium discoideum* suggests the atypical MAPK Erk2 downregulates the function of the cAMP-specific phosphodiesterase RegA to regulate progression of the developmental life cycle. A putative MAPK docking motif located near a predicted MAPK phosphorylation site was characterized for contributions to RegA function and binding to Erk2 because a similar docking motif has been previously characterized in the mammalian PDE4D phosphodiesterase. The overexpression of RegA with alterations to this docking motif (RegA^{D-}) restored RegA function to *regA*⁻ cells based on developmental phenotypes, but low-level expression of RegA^{D-} from the endogenous *regA* promoter failed to rescue wild-type morphogenesis. Co-immunoprecipitation analysis indicated that Erk2 associates with both RegA and RegA^{D-}, suggesting the docking motif is not required for this association. Epistasis analysis between *regA* and the only other *Dictyostelium* MAPK, *erk1*, suggests Erk1 and RegA can function in different pathways but that some *erk1*⁻ phenotypes may require cAMP signalling. These results imply that MAPK downregulation of RegA in *Dictyostelium* is accomplished through a different mechanism than MAPK regulation of cAMP-specific phosphodiesterases in mammalian cells and that the regulation in *Dictyostelium* does not require a proximal MAPK docking motif.

INTRODUCTION

Many signalling pathways make use of cAMP as a second messenger to regulate cellular activities within the cell [1–3]. Levels of cAMP are typically modulated through adenylyl cyclase and phosphodiesterase activities and these enzymes are often localized to complexes within cells to facilitate rapid and specific responses [4–6]. In mammals, eight different cAMP phosphodiesterase genes, each encoding several isoforms, provide a multitude of phosphodiesterases to regulate cAMP signalling in different tissues and subcellular locations, and some of these phosphodiesterases have been the targets of therapeutic inhibitors for the treatment of different clinical states [7, 8]. In addition to different isoforms, the function of some phosphodiesterases can be regulated through the phosphorylation of specific residues by upstream protein kinases, including mitogen-activated protein kinases (MAPKs) [9–14]. The complexity of phosphodiesterase expression and function in mammals presents significant challenges in understanding the function and regulation of specific phosphodiesterases in various signalling pathways [15–17]. However, simpler eukaryotes have far fewer phosphodiesterases and protein kinases and therefore offer an opportunity to better assess the regulation of specific phosphodiesterases *in vivo* using genetic approaches [18].

The soil amoeba *Dictyostelium discoideum* uses cAMP as both an extracellular and an intracellular signalling molecule [2, 19]. Starved cells release and chemotax to extracellular cAMP as a mechanism for individual cells to find neighbouring cells and form a multicellular aggregate. Extracellular cAMP continues to play a role as the aggregate undergoes morphogenesis to form a migratory slug and eventually a fruiting body with a mass of spores on top of a stalk as suggested by the presence of cell surface cAMP receptors during the multicellular stages [20–23]. Intracellular cAMP also has a role in regulating developmental progression through the activation of a cAMP-specific protein kinase (PKA) [24–27]. An important regulator of cAMP levels and PKA is the intracellular cAMP-specific phosphodiesterase RegA [28–30]. The loss of RegA results in accelerated development and precocious spore formation resulting in distorted fruiting body morphology. This aberrant development resembles the phenotypes associated with the increased PKA function, either from overexpression of the PKA catalytic subunit or from the loss of the inhibitory regulatory subunit [26, 31, 32]. RegA protein levels are highest during aggregation and early mound stage and RegA function has been associated with proper chemotactic movement [33, 34]. Histidine kinase regulation of RegA during the initiation of spore formation has also been characterized and occurs through a phospho-relay mechanism

that results in the phosphorylation of the RegA D212 residue [30, 35]. Spore differentiation signals acting through histidine kinases can downregulate RegA function during this stage of development [36].

An earlier genetic study in *Dictyostelium* has indicated that the aggregation defect of an MAPK mutant *erk2^{RE}* (reduced expression of Erk2) can be corrected by disruption of the *regA* gene, suggesting Erk2 is a negative regulator of RegA [37]. The *erk2^{RE}* mutant is incapable of generating sufficient cAMP and the loss of RegA rescues this defect by allowing higher levels of cAMP [38]. A phospho-ablative alteration of the T676 residue of RegA that lies within a conventional MAPK substrate recognition sequence (P/XXS/TP) prevented the rescue of cAMP levels, suggesting that Erk2 downregulates RegA through the phosphorylation of this residue [37]. Phospho-ablative and phospho-mimetic alterations at this site are consistent with a negative regulation of RegA function through phosphorylation at this residue [39]. This phosphorylation site occurs in a region near the carboxyl terminus that corresponds to a similar region in which MAPK phosphorylation regulates the mammalian cAMP-specific phosphodiesterase PDE4D [12]. In most isoforms of PDE4 phosphodiesterases this phosphorylation event reduces the function of the phosphodiesterase. Subsequent characterization of PDE4 isoforms also revealed a MAPK docking motif near the targeted phosphorylation site and an alteration of this motif reduced MAPK interaction and negative regulation [12]. The similarity in MAPK regulation of phosphodiesterase function in *Dictyostelium* and mammals suggests the possibility of conserved regulatory mechanisms of phosphodiesterase function in diverse eukaryotes.

The *Dictyostelium* Erk2 shares sequence similarity to a group of atypical MAPKs that have been conserved in most eukaryotes except for fungi [40, 41]. Erk2 activation occurs in the absence of MekA (also known as Mek1), the sole conventional MAPK kinase in *Dictyostelium* and this MAPK-independent regulation is also shared by atypical MAPKs in other eukaryotes [40]. Erk2 is activated in response to chemotactic stimulation, either by folate or by cAMP, and Erk2 function is required for chemotactic movement during foraging and the aggregation phase of multicellular development. A recent phosphoproteomic analysis of Erk2 revealed a strong preference for substrates with sequence S/TPR/K but genetic evidence suggests that this consensus sequence is less restrictive [39, 41, 42]. Mutational analysis of EppA (Erk2 phosphorylated protein) and RegA indicates that MAPK substrate sites without a positively charged residue after the proline position are critical regulatory sites *in vivo* [39, 42]. In addition to Erk2, the *Dictyostelium* genome encodes only one other

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Abbreviations: AKAP, A-kinase-activated protein; MAPK, mitogen-activated protein kinase; PKA, protein kinase A.

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Two supplementary figures are available with the online version of this article.

MAPK, Erk1, and this MAPK has greater sequence similar to more typical MAPKs that can be activated by conventional MAPK kinases [40, 43, 44]. Cells lacking Erk1 form small aggregates when starved and these aggregates have accelerated but relatively normal morphogenesis. Overexpression of Erk1 results in normal sized aggregates that are delayed in the transition to form migratory slugs. Erk1 is activated in response to chemotactic stimulation but this activation occurs as a secondary response that requires both Erk2 and MekkA function [45].

In this study a putative MAPK docking site was identified in RegA and characterized for its potential role in the docking of Erk2 and the downregulation of RegA function. Developmental phenotypes of *regA* mutants with alterations in this putative docking site were examined to determine how these alterations impacted RegA function. The interactions between RegA and Erk2 *in vivo* were also examined through the analysis of immunoprecipitates. In addition to analysing the Erk2–RegA interactions, epistatic relationships between *regA* and *erk1* mutations were analysed to assess if Erk1 might also be a potential regulator of RegA.

METHODS

Strains and culture conditions

All *Dictyostelium* strains were axenic and isogenic except for the loci noted. The parental strain KAx3 (also called Ax3 or Ax4) and a thymidine auxotroph, JH10, were used for creating *regA*[−] strains through a partial knockout of the *regA* locus with a blasticidin resistance gene as previously described [28, 30, 39, 46]. Cells were grown in axenic HL-5 medium with or without supplementary thymidine [47–49]. Cells were electroporated and selected in appropriate medium containing either G418 or blasticidin and multiple clonal transformants were isolated [50]. Strains were also grown on bacterial lawns and the developmental phenotypes were assessed in the plaques cleared by the *Dictyostelium* growth [46]. Levels of RegA were determined with immunoblots probed with antiserum generated against a RegA peptide (PSSHRVSDFSDEYSPC) located near the amino terminus or a RegA peptide (CKSKLPKIDEEENR) located near the carboxyl terminus (underlined residues were used for peptide attachments to adjuvants or to matrix) [39]. Erk2 detection on immunoblots used antiserum against an Erk2 peptide (ERKKQTNPTKPD $\underline{C}$) located near the carboxyl terminus.

Recombinant DNA methods

Alterations of the putative RegA D-motif were created through PCR mutagenesis of the *regA* ORF. Briefly, the codons for arginines, leucines and isoleucine in the motif (residues 588–596) were changed to alanine using the oligonucleotides listed in Fig. S1 (available in the online version of this article). The mutated *regA* gene, *regA*^{D−}, was sequenced to verify changes and then inserted as a *Bam*HI/*Xba*I fragment into the previously described *Dictyostelium* expression vectors pTXGFP and pDXA-3H [51, 52]. For creating the *regA* allele knock-in mutants the wild-type *regA* and *regA*^{D−} ORFs were

inserted as a *Bam*HI/*Xba*I fragment into a pBluescriptSK- (Stratagene) vector with a *Bam*HI/*Sal*I fragment of the *thyA* gene previously inserted into the *Bam*HI/*Xho*I sites of the multiple cloning site region. The knock-in vectors were linearized at a unique *Sal*I site near the amino terminal end of the *regA* ORF and transformed into *regA*[−] *thyA*[−] cells. The single crossover recombination event into the *regA* locus was verified by PCR analysis as previously described [39]. The knock-in vectors lacked promoter sequences for *regA* ORFs and therefore only the knock-in allele adjacent to the endogenous promoter was expressed.

Dictyostelium development and phenotype analysis

Dictyostelium strains were grown in HL-5 medium shake culture to mid-logarithmic phase ($\sim 3 \times 10^6$ cells ml^{−1}) and then collected by centrifugation. Cells were washed in phosphate buffer (12 mM NaH₂PO₄ adjusted to pH 6.1 with KOH) and typically suspended at 3×10^7 cells ml^{−1} before plating on non-nutrient agar (1.5% in phosphate buffer) plates for development. Images of developmental morphology were recorded at the times indicated using a Nikon SMZ-U dissecting microscope with a transmitted light source.

Immunoprecipitation and Western blot

For immunoprecipitation, cells were grown in HL-5 medium shaking cultures until they reached $\sim 5 \times 10^6$ cells ml^{−1}. The cells were harvested, washed and suspended at 5×10^7 cells ml^{−1} in phosphate buffer. The cells were starved for 6 h in a shaking culture with 100 μ M cAMP stimulation every 15 min during the last 3 h to induce cAMP responsiveness. Cells were shaken 15 min after the last pulse of cAMP and then stimulated a final time with 400 μ M cAMP. Cells samples were collected at the indicated times by transferring to ice-cold phosphate buffer on ice. Cells were concentrated by centrifugation and cell pellets were frozen at -80°C . Cell pellets were suspended in 1% (v/v) Triton X-100 in immunoprecipitation buffer composed of 50 mM Tris-HCl pH 8, 150 mM NaCl, 10 μ g aprotinin ml^{−1}, 10 μ g leupeptin ml^{−1} and 1 mM PMSF. The cell supernatant obtained after centrifugation (1000 g for 10 min at 4°C) was incubated with anti-RegA antiserum at 4°C overnight with continuous rocking. Protein-A agarose beads were added to the samples and rocked for another 2 h. The beads were pelleted, washed multiple times in immunoprecipitation buffer and wash buffer (50 mM Tris-HCl pH 8 and 150 mM NaCl) and then treated with 5 $\times$ SDS-PAGE gel loading buffer and heated. Proteins in the samples were subjected to SDS-PAGE and immunoblot analysis. Immunoblotting and band detection using chemiluminescence was conducted as previously described [45].

RESULTS

Putative MAPK docking sites in RegA

Earlier studies have indicated that the mammalian MAPK Erk2 can bind PDE4D phosphodiesterases through a docking motif near the reported MAPK phosphorylation site [12, 14]. A similar interaction might be expected for the MAPK

	++XXXXHXH
DdRegA	I L E G L N E D Q Y K E L R R S V V Q L I L A T D M Q N H F E H T N K F Q H H L N N L P F D R N - - - - - K K E D R Q
DpRegA	I L E G L N E D Q Y K E L R R S V V Q L I L A T D M Q S H F E Y I N K F Q H H L N N L P F D R N - - - - - K K E D R Q
TlRegA	I L E G L N E D Q Y K E L R R S V V Q L I L A T D M A N H F E H I S K F Q H H L N S S E F D R N - - - - - K K E D R Q
CfRegA	I L E G L N E D Q Y K E L R R S V I Q L I L A T D M S F H F E Y I N K F Q H H L N N Q P F D R N - - - - - K K E D R Q
AcRegA	I L T G L T D L E Y R E L R K I V V N C I L S T D M L K H V E I V T K F T T M V D Q - - F T R E - - - - - N R E H R A
HsPDE4	I F Q N L T K K Q R Q S L R K M V I D I V L A T D M S K H M N L L A D L K T M V E T K K V T S S G V L L L D N Y S D R I
	++XXXXXXHXH
	P X T P
DdRegA	M I L N F L I K C G D I S N I A R P W H L N F E W S L R V S D E F F Q Q S H Y E T I C G Y P V T P F M D K T K T T R A R
DpRegA	M I L N F L I K C G D I S N I A R P W H L N F E W S L R V S D E F F Q Q S H Y E T I C G Y P V T P F M D K T K T T R A R
TlRegA	Q I L N F L I K C G D I S N V A R P W E L N F E W S I R V S D E F F Q Q S Q Y E K V C G F P V A P F M N K T Q T T R A R
CfRegA	M I L N F L I K C G D I S N V A R P W H L N Y E W S I R V S D E F F Q Q S N F E K V C G Y P V T P F M D K T K T T R P R
AcRegA	L L L E I I L K C A D I S N P T R P P R I A A Y W S Q M V Q E E F F A Q G D K E K E G L P V S P F M D R D N S L P G K
HsPDE4	Q V L Q N M V H C A D L S N P T K P L Q L Y R Q W T D R I M E E F F R Q G D R E R E R G M E I S P M C D K H N A S V E K
DdRegA	I A A D F I D F V A S P L F Q S M A K F L K E S Q F L L K V I S K N R E N W Q A Y M E L Q K E G K C N D D D - - - - -
DpRegA	I A A D F I D F V A S P L F Q C M A K F L G E S K F L L K V I S Q N R E N W Q A Y M E L Q K E G K L N D D E - - - - -
TlRegA	I A A D F I D F V A L Q L Y Q A L A Q F L V E S K I L D I M S K N R Q N W Q A I I E L Q K E G K Y K D D L - - - - -
CfRegA	I A A D F I D L V A L P L F Q N L C R F L T E A K V L M E M M S E N R H K W Q I E L Q K L E G R K E G D E Q K A K D N D
AcRegA	M V V G F V D F F V A P L Y A T L V K L L P G M K S C H D R L L D N R A R F A L L A Q E A E D K K K A A T T - - - - -
HsPDE4	S Q V G F I D Y I V H P L W E T W A D L V H P D A Q D I L D T L E D N R E W Y Q S T I P Q S P S P A P D D P E E G R Q G
	P X S P
DdRegA	- - - - - L Q F M E D P T I L V K S K L P K I D E E E -
DpRegA	- - - - - L Q Y Y E N P A L L V N S K L P K I D E E A -
TlRegA	- - - - - Y F I K E P T L M V Q N Q K L A K I N E C D -
CfRegA	S P T T P T T N L N S S S S S K S A D I K L P S I D E E E -
AcRegA	- - - - - T T T A A A A N Q G G S G S P A T P A S S -
HsPDE4	Q T E K F Q F E L T L E E D G E S D T E K D S G S Q V E E D -
	F Q F

Fig. 1. Alignments of *Dictyostelium discoideum* RegA (DdRegA) putative docking motifs (yellow shading) and MAPK phosphorylation sites (blue shading) and flanking residues with other dictyostelids (Dp, *Dictyostelium purpureum*; Tl, *Tieghemostelium lacteum*; Df, *Cavenderia fasciculata*), *Acanthamoeba castellanii* (Ac) and human (HsP4D3 – PDE4D3 isoform) cAMP-specific phosphodiesterases. Sequences correspond to residues 575–764 of DdRegA. Consensus sequences are listed above or below the alignments: docking sites ($+_{1-2}X_{4-6}HXH$) and (FQF) and phosphorylation sites (PXS/TP) where the + represents a positively charged residue, H represents a large hydrophobic residue and X any residue.

regulation of the *Dictyostelium* phosphodiesterase RegA given that the predicted phosphorylation site, residue T676, is located in the same general region of the phosphodiesterase [37]. To examine the possibility that MAPKs can dock with RegA, a search for putative MAPK docking sites was conducted on RegA. MAPK docking sites typically consist of a $K/R_{(1-2)}X_{(4-6)}I/LXI/L$ motif (D-motif) and in some cases an additional FFX motif exists [53, 54]. RegA has two potential D-motifs, but only one motif (residues 588–596) resides near the MAPK phosphorylation site in the C-terminal region (Fig. 1). Alignment of *Dictyostelium* RegA with mammalian PDE4D indicates the carboxyl-terminal putative D-motif is conserved in its consensus sequence in multiple species, including other dictyostelids. However, the previously characterized D-motif in mammalian PDE4D is approximately 20 residues further toward the carboxyl terminus and not conserved in dictyostelids [12]. An FFX motif was also identified proximal to the D-motif in PDE4D but such a motif is not present in the RegA homologues in dictyostelids. The signature residues of the D-motifs in RegA or mammalian PDE4D do not include any of the residues highly conserved

among phosphodiesterases involved with metal or AMP binding, suggesting that these residues are not likely to be involved with catalytic activity [55].

Overexpression of *regA* with altered D-motif delays development

To investigate the potential role of the C-terminal D-motif in RegA, site-directed mutagenesis was used to replace the arginine and leucine residues of the motif with alanine residues. The resulting *regA*^{D−} allele was inserted into both extrachromosomal and integrating expression vectors downstream of the constitutively active *act15* promoter and introduced into *regA*[−] cells. Expression of the mutant RegA^{D−} delayed developmental progression similar to that observed for the expression of wild-type RegA from the same vectors (Fig. 2a). The extent of the delay correlated with RegA^{D−} or RegA protein levels, consistent with excessive RegA function and decreased cAMP levels as previously described (Fig. 2b). Attempts to generate clonal transformants with levels of RegA^{D−} or RegA expression similar to wild-type strains using reduced drug selection

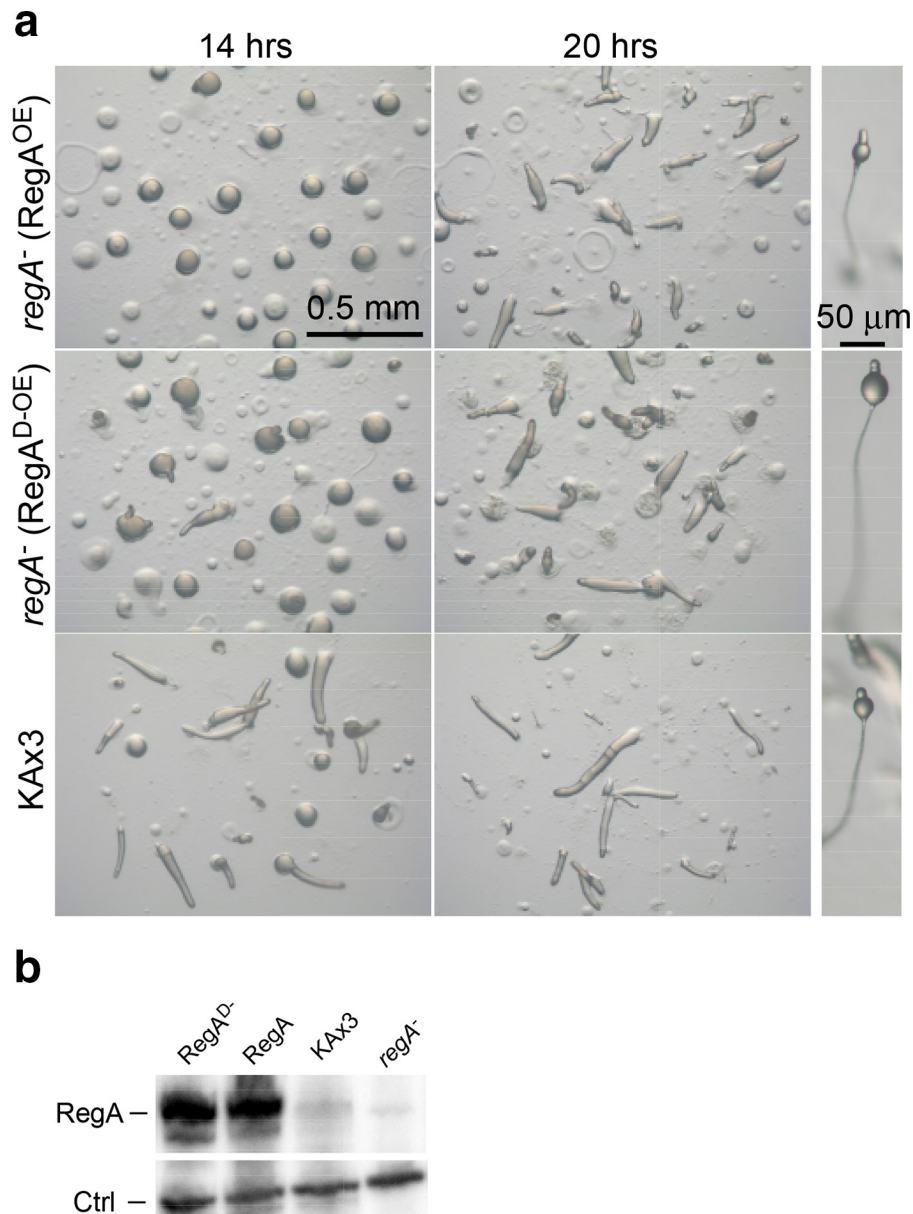


Fig. 2. Overexpression of RegA and RegA^D. (a) Developmental morphology of RegA overexpression strains. Clonal transformants of *regA*⁻ cells overexpressing wild-type RegA (RegA^{OE}) or RegA^D (RegA^{D-OE}) and parental wild-type KAx3 cells were plated for development on non-nutrient agar plates and monitored at the indicated times for developmental morphology. All images in a column correspond to the same magnification and period of starvation. Representative fruiting body structures after more than 24 h of starvation are shown in the far right column. (b) Immunoblot of RegA and RegA^D strains. Cells were harvested after 4 h of starvation, lysed and subjected to immunoblot analysis using antiserum generated against a peptide sequence in the amino terminal region of RegA (upper panel). Each lane was loaded with lysates from 1×10⁶ cells. Non-specific background band was used as a loading control (Ctrl – lower panel).

(2 μg ml⁻¹ G418) was not successful, suggesting that overexpression of these genes is primarily attributed to the strong *act15* promoter. The antiserum against RegA also detects a non-specific band in the lysates of clonal *regA*⁻ (knockout) strain lysates that runs slightly lower than RegA in wild-type lysates.

Expression of RegA^D from the endogenous promoter does not rescue RegA function

To assess the role of the putative D-motif under conditions of lower RegA expression, vectors containing promoter-less *regA* alleles were integrated into the disrupted *regA*⁻ locus through a single crossover mechanism to allow a single copy

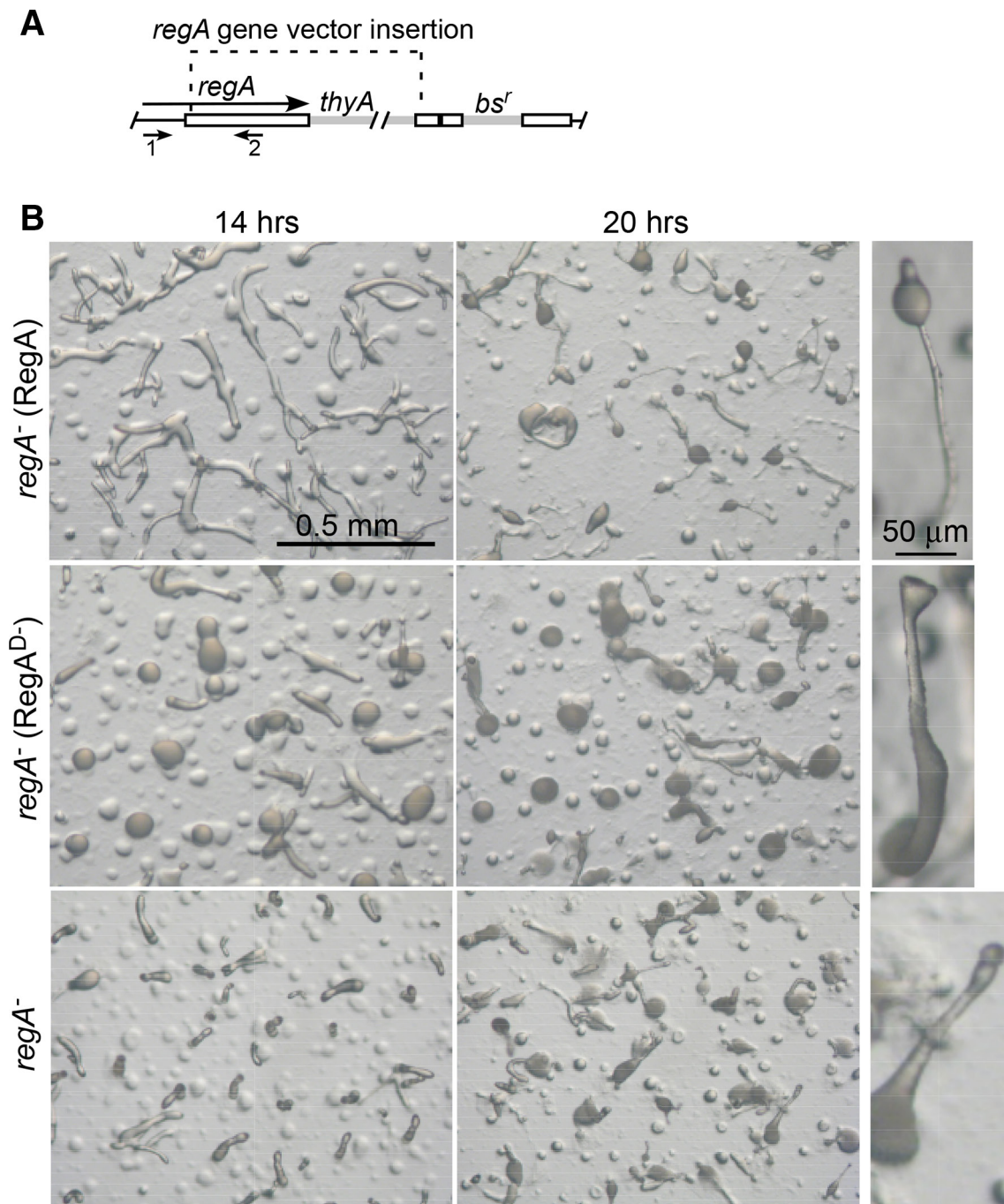


Fig. 3. Integration and expression of *regA* knock-in alleles. (a) Map of the *regA* knock-in allele in the disrupted *regA* locus through a single crossover integration. *regA* ORF (open bar) and intron (black bars) sequences are indicated along with the replacement of *regA* sequences with the blasticidin resistance gene (*bs^r* – grey bar) in the original knockout allele. The dotted line indicates the borders of the integration vectors used for the *regA* allele knock-ins. The grey bar internal to the vector represents the *thyA* gene and vector backbone of integration constructs (internal slashes indicate approximately 4 kb of sequences not represented in the map). The black line represents sequences upstream and downstream of the coding region and the arrow indicates *regA* transcription from the endogenous promoter. (b) Developmental morphology of *regA*⁻ cells with knock-in alleles expressing either wild-type RegA or RegA^{D-}. The morphology of *regA*⁻ cells without knock-in alleles is shown for comparison. Cells were plated from suspensions of 5×10^7 cells ml⁻¹ and images were recorded at the times indicated above columns. The far right column displays final fruiting body morphology.

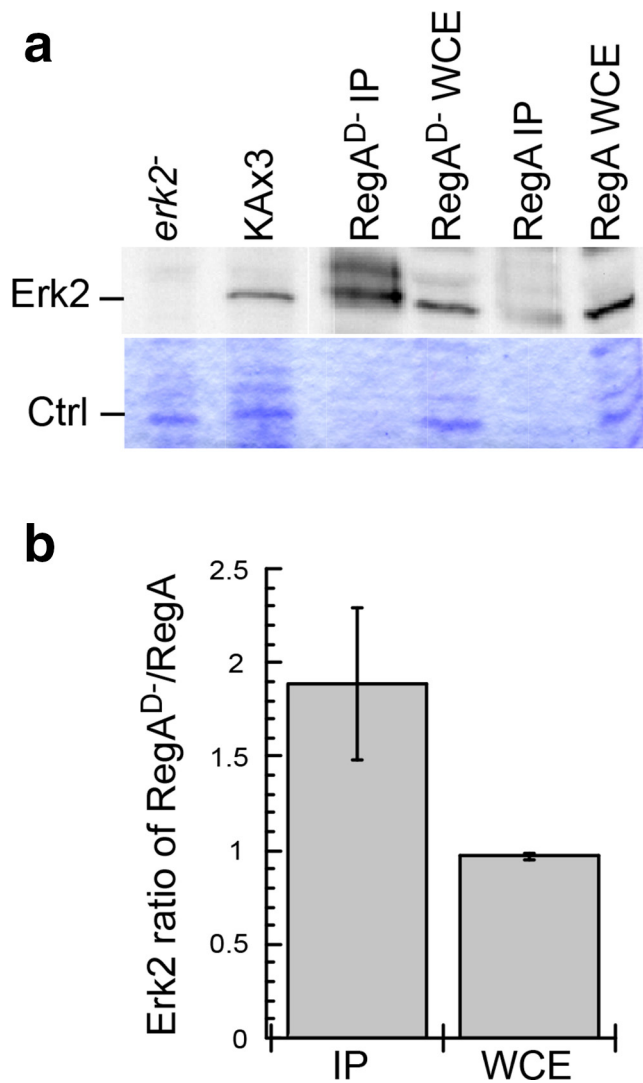


Fig. 4. Erk2 associated with RegA immunoprecipitates. *regA*⁻ cells with knock-in alleles expressing RegA^{D-} or wild-type RegA were starved for 6 h and pulsed with cAMP as described in the Methods. (a) Immunoprecipitates of RegA using antiserum against the amino-terminal side RegA peptide were separated by SDS PAGE and immunoblotted for Erk2 detection using the Erk2 antiserum (lane indicated by IP). Whole cell extracts (WCE) separated prior to immunoprecipitation were included along with whole cell extracts of parental strain KAx3 and an *erk2*⁻ knockout strain. The blot is representative of three blots of two independent experiments. Coomassie staining of the blotted gel (lower panel) was used as a loading control (Ctrl). (b) Ratios of lower Erk2 band intensity in RegA^{D-} to RegA immunoprecipitates (IP) and whole cell extracts (WCE) were quantified using ImageJ and averaged for two independent immunoprecipitations. Error bars indicate range of values.

of the *regA* allele to be expressed from the endogenous *regA* promoter (Fig. 3a). This strategy has been previously used to assess other *regA* alleles [39]. Expression of wild-type RegA in the gene knock-in strains resulted in normal fruiting body morphology without a delay in development, suggesting the strain produced sufficient but not excessive levels of RegA (Fig. 3b). Development was slightly accelerated (culmination at 22 h of starvation) but not as much accelerated as

that associated with *regA*⁻ cells (culmination at 18 h). Also, *regA*⁻ cells do not undergo normal fruiting body morphogenesis during culmination but rather produce a stalk with most spores located at the base. Therefore, all subsequent analyses and comparisons of RegA and RegA^{D-} function were conducted in these gene knock-in strains. The development of RegA^{D-} knock-in cells resulted in small aggregates, precocious spore formation and aberrant fruiting body morphology similar to *regA*⁻ strains. This phenotype suggests that RegA^{D-} does not provide sufficient RegA function when expressed from the endogenous *regA* promoter. The level of RegA^{D-} and RegA expressed from the endogenous *regA* promoter were comparable at 4 h of development, suggesting that the absence of RegA function in RegA^{D-} mutants is probably not due to general lack of protein stability (Fig. S1d).

Putative D-motif is not essential for Erk2 and RegA interactions

The reduction of RegA function of RegA^{D-}-expressing cells was not expected based on the assumption that this putative D-motif facilitates an interaction with a negative regulator such as Erk2. However, if Erk2 or other proteins interacted with this motif then these associations might also coordinate complexes with positive regulators of RegA. To investigate the possibility that Erk2 associates with RegA and also the potential importance of the D-motif, immunoprecipitates of RegA^{D-} and RegA protein complexes from *regA*^{D-} and *regA* knock-in strains (*regA*⁻ background) were assessed for the presence of Erk2. Lysates of cAMP-stimulated cells from different intervals of stimulation (0, 40 and 70 s post-stimulation) were combined for RegA immunoprecipitations because possible interactions between RegA and Erk2 could be transient over the course of the cellular response. Immunoblots of RegA^{D-}- and RegA-expressing cells displayed Erk2-specific bands in whole cell lysates and in both RegA and RegA^{D-} immunoprecipitates, indicating that Erk2 can associate with RegA *in vivo*, regardless of the D-motif alteration (Fig. 4a). Unexpectedly, the intensity of the Erk2 bands was significantly higher in the immunoprecipitates of RegA^{D-} compared to RegA, suggesting that the Erk2 association with RegA is increased by the alteration of the D-motif (Fig. 4b). Levels of Erk2 in whole cell extracts were comparable in both RegA^{D-}- and RegA-expressing cells, suggesting the increased association of Erk2 with RegA^{D-} was not due to increased Erk2 expression. Interestingly, the presence of a slower migrating, possibly an Erk2-specific, band was more apparent in the RegA^{D-} immunoprecipitate and whole cell extract compared to those of the RegA cells. The identity of this protein is unknown but its increased presence in RegA^{D-} cells and immunoprecipitates might be related to the alteration of the D-motif.

Comparison of Erk1 and RegA developmental phenotypes

Based on the co-immunoprecipitation analysis of RegA and Erk2, the D-motif is not essential for Erk2 association with

RegA but could potentially be involved with other MAPK interactions. The only other *D. discoideum* MAPK is Erk1 and potential relationships between Erk1 and RegA function during development have not been investigated. To assess whether Erk1 and RegA might function in the same pathway, the developmental phenotypes of *erk1* and *regA* mutants were analysed and compared. The loss of either *regA* or *erk1* accelerates development but with different morphologies (Fig. 5a). The *erk1*⁻ cells form small aggregates with accelerated tip formation, whereas *regA*⁻ cells form larger aggregates similar in size to that of wild-type cells. The *erk1*⁻ aggregates develop into small fruiting bodies with wild-type morphology but *regA*⁻ aggregates form aberrant fruiting bodies with many spores located at the base of the stalk. Double *erk1*⁻*regA*⁻ mutants form smaller aggregates than *erk1*⁻ cells and these aggregates develop into small fruiting bodies with the aberrant morphology associated with *regA*⁻ mutants. This combination of *erk1*⁻ and *regA*⁻ phenotypes in the double mutant suggests RegA and Erk1 can operate in separate developmental signalling pathways. The impact of overexpressing RegA or Erk1 was also examined by using the *act15* promoter to drive expression of either gene from an extrachromosomal vector. Overexpression of RegA in *erk1*⁻ cells increased aggregate size, suggesting that excessive RegA function can partially mask the small aggregate phenotype associated with *erk1*⁻ mutants (Fig. 5b). Likewise, overexpression of Erk1 delayed the progression of *regA*⁻ aggregates into slugs (Fig. 5c). The ability of overexpressed Erk1 or RegA to mask the null phenotype of the other suggests that Erk1- and RegA-mediated pathways can directly or indirectly impact each other.

Erk1 overexpression does not correct aberrant developmental morphology of Reg^{D-} cells

Although RegA and Erk1 can function in separate pathways, this does not necessarily exclude the possibility that these proteins might interact and also facilitate a common pathway. As a MAPK, the Erk1 could possibly interact with the putative carboxyl-terminal D-motif in RegA and the overexpression of Erk1 might potentially compensate for alterations of this motif. Overexpression of Erk1 resulted in a slight delay in the progression of mounds to form tips in both RegA- or RegA^{D-}-expressing cells (Fig. 6). With or without overexpression of Erk1, the RegA^{D-} knock-in cells displayed the large mass of cells at the base of the fruiting body, indicating that overexpression of Erk1 cannot correct the aberrant fruiting body morphology associated with the altered motif. Overexpression of Erk1 also did not affect the wild-type morphology of cells expressing the wild-type *regA* allele. These results suggest that increased Erk1 expression cannot correct for alterations in the RegA D-motif, at least with respect to morphogenesis of the fruiting body structure.

DISCUSSION

MAPK regulation of phosphodiesterase function is supported by biochemical evidence in mammals and genetic evidence

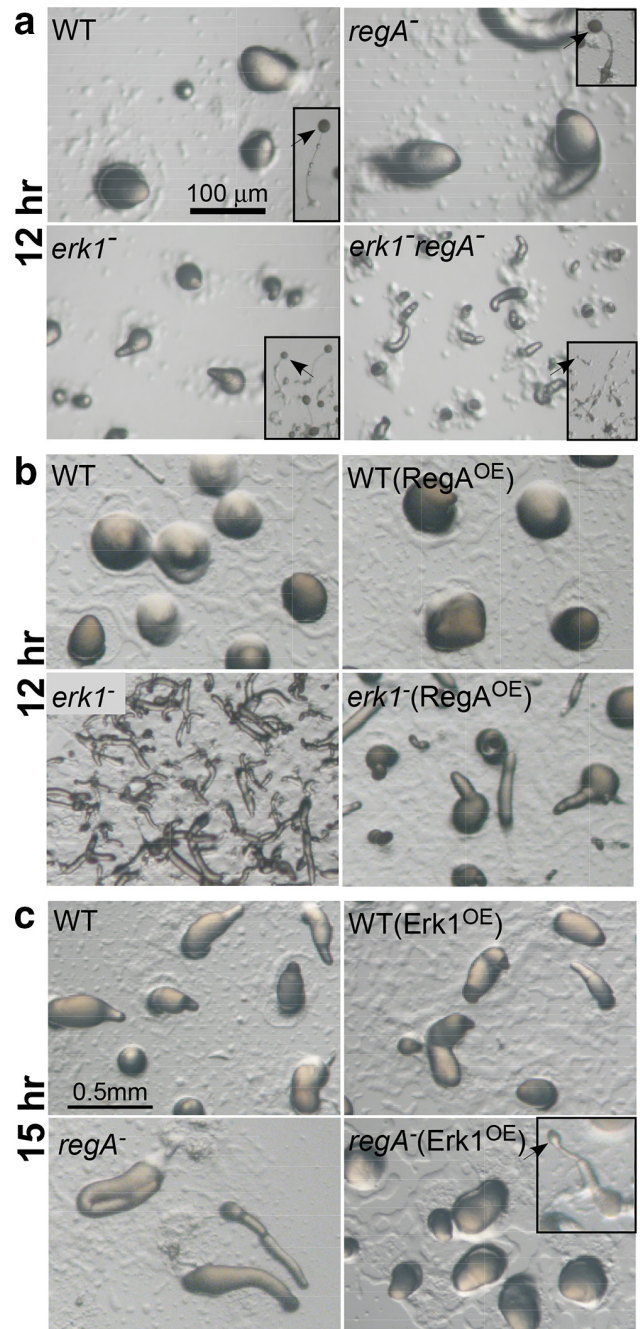


Fig. 5. Developmental morphology of strains lacking or overexpressing RegA and Erk1 on non-nutrient plates. Cells were grown and prepared for development as described in the Methods. (a) Parental KAx3 (WT), *regA*⁻, *erk1*⁻ and *erk1*⁻*regA*⁻ (double mutant) cells were developed for 12 h. Insets show representative mature fruiting bodies (1/3 magnification). (b) KAx3, KAx3 overexpressing RegA (RegA^{OE}), *erk1*⁻ and *erk1*⁻ overexpressing RegA cells were developed for 12 h. (c) KAx3, KAx3 overexpressing Erk1 (Erk1^{OE}), *regA*⁻ and *regA*⁻ overexpressing Erk1 were developed for 15 h. Inset shows maturing fruiting body structure at the same magnification. All images, except some insets, in each panel are the same magnification and cells were developed side-by-side with the same cell densities. Arrows indicate spore mass on top of the stalk.

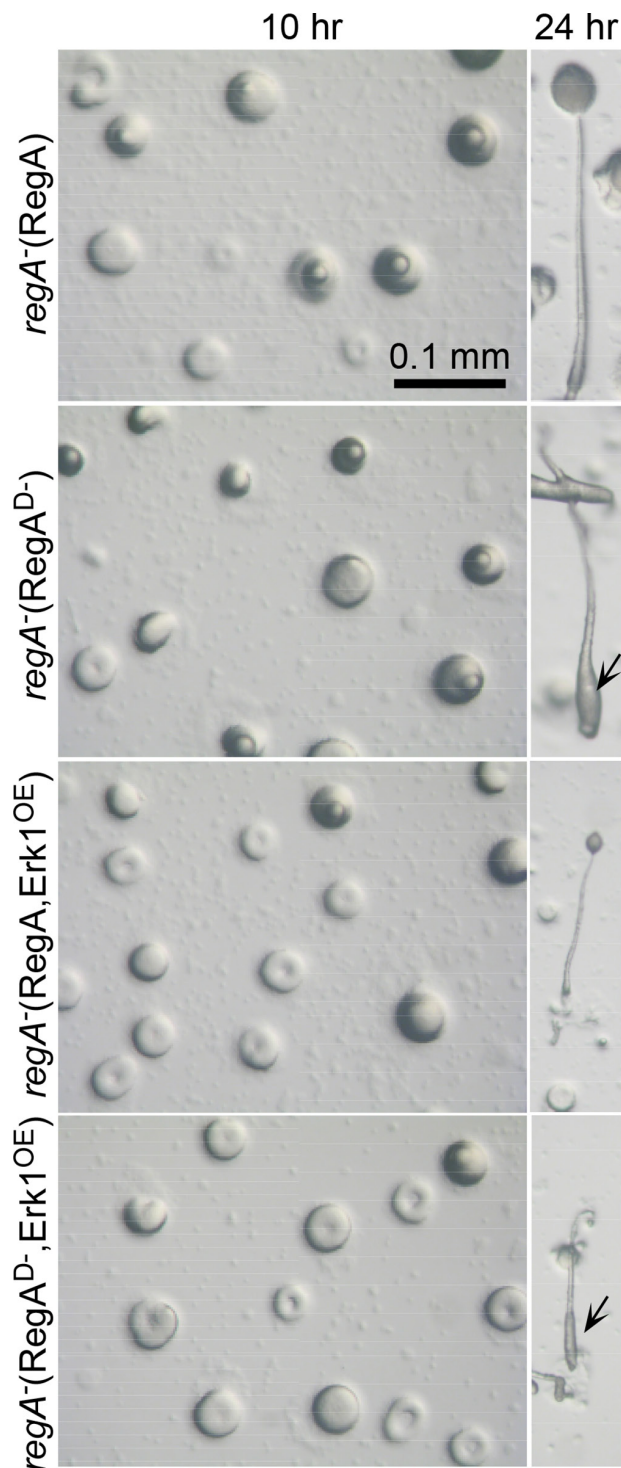


Fig. 6. Developmental morphology of RegA and RegA^{D-/-} cells with Erk1 overexpression. *regA*^{-/-} cells with knock-in alleles expressing either RegA or RegA^{D-/-} and with or without Erk1 overexpression (Erk1^{OE}) were grown and developed on non-nutrient plates for the times indicated in each column as described in the Methods. All images in each panel are at the same magnification and developed side-by-side with the same cell densities. Arrows indicate large stalk base due to aberrant spore localization.

in *Dictyostelium* and so the possibility of a conserved regulatory mechanism was plausible given that the MAPK phosphorylation occurs near the carboxyl terminus in both cases. Conserved docking motifs on MAPK substrates in many organisms suggest that docking through these sites is a typical mechanism for recognition and phosphorylation of substrates. The presence of a putative docking motif near the predicted phosphorylation site on RegA further supported a docking mechanism for Erk2 regulation of the phosphodiesterase, but the results of this study suggest that this putative docking motif is not required for Erk2 interactions. The developmental phenotypes of cells expressing RegA^{D-/-} from the endogenous *regA* promoter are consistent with a loss of RegA function as opposed to the gain of RegA function that might be expected through the impairment of a negative regulation mechanism. Perhaps the alignment shift in the putative docking motif and phosphorylation site in RegA with those found in the mammalian PDE4 phosphodiesterase reflects a significant difference in how MAPKs and phosphodiesterases interact. Unlike the mammalian PDE4 phosphodiesterase, the amoeba RegA phosphodiesterases do not have other putative docking motifs in the carboxyl terminal region.

The characterization of RegA immunoprecipitates revealed that Erk2 does associate with RegA *in vivo*, consistent with Erk2 phosphorylation and negative regulation of RegA. Alteration of the putative D-motif of RegA enhanced rather than destroyed this association, suggesting Erk2 associates with RegA through other docking motifs or some other mechanism (Fig. 7). Another putative MAPK docking motif exists on RegA but this motif is located much closer to the amino terminus and lacks a positively charged residue at the second residue position. This second motif is present in other dictyostelids but is absent in the orthologues of other amoeba and mammals (Fig. S2). This second motif is much further away from the predicted Erk2 phosphorylation site (T676) but the possibility that these two sites could be brought into close proximity through folding of the RegA polypeptide chain cannot be excluded. Alternatively, Erk2 might not recognize a conventional MAPK docking site because Erk2 belongs to a group of atypical MAPKs and it remains to be determined whether atypical MAPKs recognize the same or different docking motifs as do typical MAPKs. Erk2 could also be indirectly associated with RegA through some type of scaffolding mechanism. Mammalian phosphodiesterases are often associated with AKAPs (A-kinase-associated proteins) that allow for the regulation of cAMP-dependent protein kinase activity [56]. While AKAPs appear to be specific to animals, *Dictyostelium* and other organisms might have analogous scaffolding proteins that facilitate phosphodiesterase interactions [57].

When expressed from the endogenous *regA* promoter, the RegA^{D-/-} did not rescue RegA function required for normal fruiting body morphology whereas wild-type RegA did rescue this morphology. Both RegA^{D-/-} and RegA provided excess RegA function when overexpressed from a heterologous promoter, suggesting the alteration of the docking motif does not destroy RegA function. However, the possibility

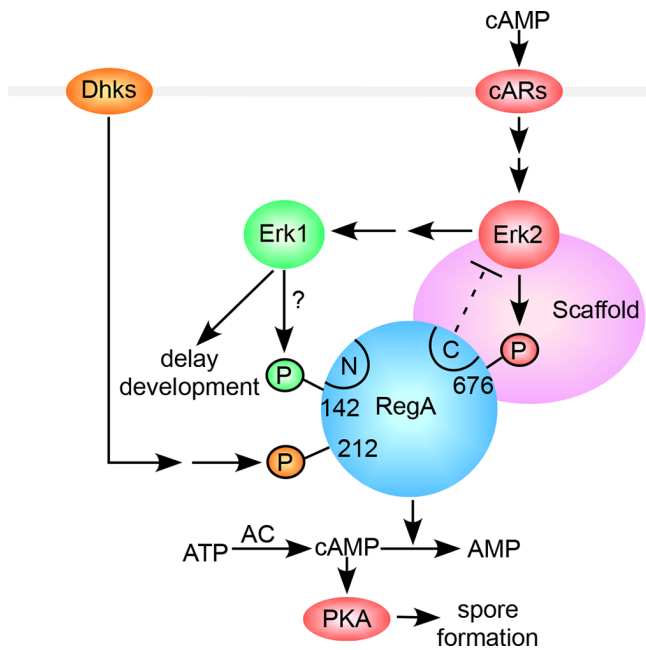


Fig. 7. Model of MAPK regulation of RegA. Stimulation of cAMP receptors (cARs) leads to Erk2 activation and RegA phosphorylation on residue T676 (negative regulation). RegA contains two putative MAPK docking motifs, one near the amino terminus (section designated N) and another near the carboxyl terminus (section designated C). Erk2 association with RegA does not require, and may be inhibited by, the carboxyl-terminal docking motif. Erk2 interactions with RegA might require a docking site distinct from those of conventional MAPKs or Erk2 might interact indirectly with RegA through another protein (e.g. a scaffold). Erk1 is activated downstream of Erk2 and Erk1 might phosphorylate RegA on residue S142 (positive regulation). Erk1 can function in pathways independent of RegA. Histidine kinase pathway signalling also leads to the phosphorylation of RegA. RegA function reduces cAMP levels and PKA catalytic activity that promote spore formation. Spatial proximity between the amino and carboxyl terminus is unknown. Double arrows represent multiple unknown signalling steps.

that the RegA^{D-} protein has subtle changes in protein stability or catalytic activity at some stage in development cannot be ruled out. The inability of RegA^{D-} to rescue RegA function at low expression levels could also be the result of increased RegA^{D-} association with Erk2, as suggested by the co-immunoprecipitation analysis. Enhanced interactions with Erk2 could possibly reduce the function of RegA^{D-} but when overexpressed some RegA^{D-} might exist independently of Erk2 and escape this negative regulation. An increased ratio of RegA to Erk2 might also be a contributing factor in the delayed developmental phenotype observed for overexpression of the wild-type RegA.

The role of Erk1 in the regulation of RegA remains uncertain but the ability of RegA overexpression to increase the aggregate size and slow the accelerated development of *erk1*⁻ cells suggests that increased cAMP levels are associated with *erk1*⁻ phenotype. However, Erk1 must regulate pathways that do not require RegA function because Erk1 overexpression can slow the development of *regA*⁻ cells. It is possible

that Erk1 functions to positively regulate RegA as cells adapt to chemoattractant stimulation because Erk1 activation occurs during the adaptation period when cAMP levels decrease [45]. Previous mapping of RegA phosphopeptides has indicated that serine 142 can be phosphorylated and that phenotypic analysis of phospho-mimetic and phospho-ablative mutants suggests the phosphorylation of this site upregulates RegA function [39]. While this phosphorylation of RegA would be consistent with the possibility of Erk1 regulation, the inability of Erk1 overexpression to suppress the RegA^{D-} aberrant developmental phenotype suggests the carboxyl-terminal docking motif is unlikely to mediate this regulation. Understanding the potential Erk1 regulation of RegA will probably require a more extensive analysis that includes Erk1 and RegA associations. In addition to Erk1, the role of other protein kinases in RegA regulation has been analysed. Histidine kinase and phosphorelay proteins facilitate the phosphorylation of RegA at residue D212 and this regulation impacts spore formation later in development [30, 35, 58]. It has not been determined whether RegA can be simultaneously phosphorylated at multiple sites or what possible impact one phosphorylation event might have on another. The regulation of RegA appears to be complex, suggesting RegA is downstream of many signalling pathways.

In conclusion, this investigation has revealed that the putative docking motif in the carboxyl-terminal region of RegA is not required for Erk2 downregulation of function. This finding contrasts with the mechanism proposed for MAPK regulation of phosphodiesterases in mammals and several distinctions exist between the two systems besides the docking motif [12]. In *Dictyostelium*, the phosphodiesterase RegA is negatively regulated by Erk2, an atypical MAPK and it remains to be determined if an atypical MAPK, such as the mammalian MAPK15 (also known as Erk8), contributes to mammalian phosphodiesterase regulation or whether regulation occurs only through more conventional (typical) MAPKs such as the mammalian Erk1 and Erk2 MAPKs [12]. In addition, the analysis of MAPK–phosphodiesterase interactions in mammals was accomplished through the ectopic expression of the phosphodiesterase in cell lines and these cells could potentially have altered signalling mechanisms compared to that of normal tissues. While future studies might provide insight into the apparent differences in *Dictyostelium* and mammalian phosphodiesterase regulation, the ability of MAPKs to regulate phosphodiesterase function in very diverse organisms suggests that some mechanisms of cAMP signalling may have been conserved.

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Author contributions

N.A. contributed the majority of the investigation and J.A.H. and N.A.K. also contributed to the investigation. N.A. and J.A.H. prepared the original draft and all authors contributed to the review and editing.

Conflicts of interest

The authors declare that there are no conflicts of interest.

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